

FANGDA YE

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RESEARCH INTERESTS

My work so far builds and evaluates **deep research agents**: multimodal search, grounded long-form generation, and process-level evaluation of agent trajectories. I have also worked on physically consistent video generation and test-time training for robot memory. I am interested in how agents learn to reason and act: **reinforcement learning for LLM agents**, **test-time verification and training**, **self-improving agents**, **latent reasoning**, and **world models for embodied agents**.

EDUCATION

National University of Singapore (NUS)

Singapore

Master of Computing (Computer Science) · GPA: 4.88 / 5.00

Aug 2024 – Jun 2026

- Academic Excellence Award (AY2025/26); Dean’s List in both semesters of AY2024/25.
- Thesis: *Proactive Demand Detection in Multi-Turn Conversations: Benchmark, Method, and Evaluation*. Advisor: Prof. Shuicheng Yan.

Central South University

Changsha, China

Bachelor of Engineering · GPA: 90.78 / 100

Sep 2019 – Jun 2023

PUBLICATIONS

* equal contribution.

Conference papers

- [C1] **MiroEval: Benchmarking Multimodal Deep Research Agents in Process and Outcome**
Fangda Ye, Yuxin Hu, Pengxiang Zhu, Yibo Li, Ziqi Jin, ..., Bryan Hooi, Bo An, Shuicheng Yan, Lidong Bing
Advances in Neural Information Processing Systems (NeurIPS), 2026 [\[arXiv\]](#) [\[Code\]](#)
- [C2] **Deep-Reporter: Deep Research for Grounded Multimodal Long-Form Generation**
Fangda Ye, Kuicai Dong, Zhifei Xie, Yuxin Hu, Yihang Yin, Shurui Huang, Shikai Dong, Chen Zhang, Jianzhu Bao, Shuicheng Yan
Annual Meeting of the Association for Computational Linguistics (ACL), Main Conference, 2026 [\[Paper\]](#) [\[arXiv\]](#)
- [C3] **Seeing the Unseen: Unified Visible-Invisible Motion for Physically Consistent Video Generation**
Mingyang Bao, Fangda Ye, Xihao Chen, Bobo Li, Xiangtai Li, Shengqiong Wu
Advances in Neural Information Processing Systems (NeurIPS), 2026
- [C4] **Doc-Researcher: A Unified System for Multimodal Document Parsing and Deep Research**
Kaicai Dong*, Shurui Huang*, Fangda Ye*, Wei Han, Zhi Zhang, Dexun Li, Wenjun Li, Qu Yang, Gang Wang, Yichao Wang, Chen Zhang, Yong Liu
The ACM Web Conference (WWW), 2026 [\[arXiv\]](#)

Under review

- [U1] **Temporal Contrastive Test-Time Training for Robotic Manipulation**
Yihang Yin*, Fangda Ye*, Shiguang Wu, Kunhao Pan, Ruiyang Xu, Ruyi Gan, Hao Wang
Under review

Technical reports

- [T1] **MiroThinker-1.7 & H1: Towards Heavy-Duty Research Agents via Verification**
MiroMind Team (contributor; Fangda Ye among 41 authors)
Technical report, 2026 [\[arXiv\]](#)
- [T2] **PASK: Toward Intent-Aware Proactive Agents with Long-Term Memory**
Zhifei Xie, Zongzheng Hu, Fangda Ye, Xin Zhang, Haobo Chai, ..., Deheng Ye, Chunyan Miao, Shuicheng Yan
Technical report, 2026 [\[arXiv\]](#)

RESEARCH EXPERIENCE

MiroMind (now Apodex)

Dec 2025 – Jun 2026

AI Scientist Intern · Lead: Lidong Bing · Mentors: Yue Deng, Lei Wang, Bin Wang

Singapore

- **Evaluation.** Led MiroEval (first author, NeurIPS 2026): 100 real-world tasks (70 text, 30 multimodal) that score deep research agents on report synthesis, factuality, and an audit of the research process itself. Found that process quality predicts the final outcome. MiroEval also served as the internal evaluation suite for **MiroThinker-1.7 & H1**.

- **Post-training for wide search.** Post-trained Qwen3.5-35B-A3B to issue and combine searches in parallel (+35% relative on WideSearch-Bench), owning the data pipeline end to end: 10K candidate tasks from 11 APIs and 8 query patterns, filtered to 2K hard seeds by self-consistency and two-model quality checks.
- **Agent framework.** Designed memory management, context management, and retry and error recovery for the MiroFlow agent framework; shipped in MiroFlow v1.7, which reached 82.7 on GAIA and 75.3 on BrowseComp-ZH.

Learning and Vision (IV) Lab, National University of Singapore

Aug 2025 – May 2026

Graduate Researcher, with Prof. Shuicheng Yan

Singapore

- Master’s thesis within the **PASK** project: *proactive demand detection*, i.e., deciding turn by turn whether an assistant listening to a conversation should step in without being asked.
- Built **LatentNeeds-Bench**: 100 real multi-turn conversations (3,936 turns) from podcast transcripts, annotated per turn with a two-level taxonomy of 11 demand types. Nine frontier LLMs reach a median of only 66.2% balanced accuracy with prompting.
- Built **InnerMind**: a context engine (sliding-window memory, user profile, prompt assembly) and a Qwen3-30B-A3B detector trained with SFT and then GRPO on a rule-plus-judge reward. It reaches 84.2% balanced accuracy, above Gemini-3-Flash (80.8%) and 25.3 points over its base model, at 1.4 s latency.

Huawei Singapore Research Center (Noah’s Ark Lab)

May 2025 – Dec 2025

AI Researcher Intern, with Dr. Kuicai Dong

Singapore

- Built **Deep-Reporter** end to end (first author, ACL 2026): an agent that writes long research reports with images, grounded in what it retrieves. It plans each report as a checklist of sections, searches text and images in separate streams with relevance filtering, and writes section by section with recurrent context management.
- Built **M2LongBench** (247 tasks in 9 domains over a fixed sandbox of 95K images and 108M text chunks) and curated 8K agentic traces. SFT on these traces raised Qwen3-32B’s overall score from 27.2 to 37.9, against 5.2 for naive RAG and 15.6 for a STORM-style agent.
- Designed the agent workflow of **Doc-Researcher** (co-first author, WWW 2026), deep research over long multimodal documents: a planner that narrows the document pool by 60–80% and picks the retrieval granularity, and a search–refine loop that runs until the evidence suffices. Removing the planner costs 5.7–8.3 points; the full system reaches 50.6%, 3.4× the best baseline.

INDUSTRY EXPERIENCE

TikTok

Jun 2026 – Present

AI Engineer, ranking models for e-commerce short videos in the main feed

Singapore

- **Training infrastructure.** Upgraded ranking-model training to user-level training (ULT), with a training speedup of more than 2×.
- **Model scaling.** Scaling up the ranking model and exploring Transformer architectures for feature fusion.
- **Agents for modeling.** Building agents that automate the modeling loop: adding features, upgrading model architecture, launching and iterating on experiments, and analyzing and attributing results.

HONORS AND AWARDS

Academic Excellence Award, School of Computing, NUS	Jun 2026
<i>For outstanding academic performance in AY2025/26; one recipient in the programme each year.</i>	
Dean’s List, School of Computing, NUS (AY2024/25 Semester 1 and Semester 2)	2025
Outstanding Intern, Huawei Singapore Research Center	2025
National Second Prize, China Postgraduate Mathematical Contest in Modeling	2023
Outstanding Graduate, Central South University	2023
Outstanding Undergraduate Thesis, Central South University	2023

ACADEMIC SERVICE

Reviewer, The Third Workshop on Long-Context Foundation Models (LCFM), NeurIPS 2026	2026
Reviewer, <i>ENGINEERING Information Technology & Electronic Engineering</i> (journal)	2026

SKILLS

LLM post-training: SFT and LoRA, RL for LLMs (GRPO, DPO), agentic trajectory curation, multi-GPU training

Agents and evaluation: multi-agent workflows (LangGraph, MCP), context and memory management, multimodal retrieval (Milvus), LLM-as-judge evaluation, benchmark construction

Recommendation: large-scale ranking models, Transformer-based feature fusion, training efficiency

Programming: Python, PyTorch **Languages:** Chinese (native), English (fluent)

REFERENCES

Prof. Shuicheng Yan	Dr. Lidong Bing	Dr. Kuicai Dong
National University of Singapore	MiroMind (now Apodex)	Huawei Singapore Research Center